Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/19/2017
PI Name:	Flynn-Evans, Erin E. Ph.D.		
Project Title:	Validation of Model Predictions of Alertness, Performance and Sleepiness		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Ames Research Center		
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PI Address 2:	Human Systems Integration Division, Code 262-4		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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COI Name (Institution):	Ploutz-Snyder, Robert Ph.D. (Universities Space Research Association, Columbia) Barger, Laura Ph.D. (Brigham and Women's Hospital)		
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	We compared the waking performance predictions from four models to actual performance data collected from three operational field studies and two laboratory studies. We found that all of the models provided reasonable estimates of group-average performance over time. There was no single model that appeared better able to predict group-average performance even under varying sleep loss, circadian misalignment, and varied countermeasure conditions. When we evaluated goodness-of-fit using all of the available data at the individual level, we found that none of the models yielded consistent significant associations with the data for all studies. Similarly, when we evaluated the time course of the predictions with individuals split into categories of best, average, and worst performance, we found that the models were not typically able to capture the average performance of the best and worst performers. Our findings suggest that these models are able to estimate general trends in waking performance and can be used to compare rosters to one another to		

Task Description:	aid in schedule building and general countermeasure application. Our findings also suggest that it is inappropriate to use these models to make individual-level performance predictions. Impact and Recommendations for Future Work 1. Using multiple models should provide better utility than relying on a single model. Integration of model predictions from all of the models into a user-friendly interface is advisable. 2. These models are not acceptable for making individual-level predictions. There are non-parametric and semi-parametric models that have been developed to predict performance at the individual level. Such models should be evaluated for use during long-duration spaceflight for situations where crewmembers will need to self-evaluate and self-plan schedules. 3. The addition of countermeasure information appears to improve performance predictions; however, few countermeasures have been modeled. NASA should ensure that laboratory and field data is collected and archived in a manner that is accessible to modeling groups for model building. It would also be prudent for NASA to include model development efforts in studies that are conducted to evaluate scenarios unique to NASA, such as modeling the Mars Sol. In addition, NASA should require the publication and sharing of equations used to build new model components, so that such features may be incorporated into multiple models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bio-mathematical models are powerful tools for fatigue management in challenging work environments such as transportation. Models are already in use in some settings such as commercial aviation as a means of evaluating flight crew schedules for potential fatigue-related risks. This is despite the fact that most models have not been rigorously evaluated and validated for the operations to which they are being applied and that many users are not fully aware of the limitations in which the results should be interpreted and applied. This evaluation is expected to provide the larger community of researchers and safety management personnel with a comprehensive understanding of the capabilities and limitations of each model.
Task Progress:	We have completed our evaluation comparing four models to five datasets. We met with the model developers several times to ensure that we were using the models appropriately. We next ran the sleep schedules for each individual in each study through each of the models and met with the modeling groups again to allow them to review our findings and provide suggestions regarding the model capabilities and our approach. Finally, we conducted a statistical analysis comparing the model predictions to the actual data collected in each study.
Bibliography Type:	Description: (Last Updated: 08/01/2025)